

# Database Browser

Topic:

Development of an Integrated Development Environment for Designing and Analyzing Relational  
Databases

Completed by:

24215 Pisarev Artem Sergeevich

24215 Vorona Mikhail Evgenievich

24215 Maltsev Kirill Konstantinovich

**Table of Contents**

Introduction.....3

Domain Glossary.....4

Functional Requirements.....5

Non-Functional Requirements.....7

System Design.....8

    Use Cases:.....8

# Introduction

Our goal is to create an intuitive, high-performance Integrated Development Environment (IDE) that empowers developers and data analysts to seamlessly design, manage, and optimize relational databases. By prioritizing SQLite support while maintaining extensibility for popular RDBMSs, the tool aims to bridge the gap between complex database administration and user-friendly visual interaction.

# Domain Glossary

RDBMS (Relational Database Management System) - A DBMS based on the relational data model (SQLite, PostgreSQL, MySQL, etc.).

SQLite - A lightweight relational DBMS that stores the entire database in a single file on disk.

Diagram - A graphical representation of information, data, processes, or relationships between objects using geometric shapes, lines, symbols, or graphs.

Overview Window - A user interface (UI) element that displays a reduced, summarized, or full version of a large object (document, image, map, dataset, or workspace) as a separate panel or window.

DDL (Data Definition Language) - A subset of SQL used to create and modify the database structure (CREATE, ALTER, DROP commands).

Database Schema - The logical structure of the database: tables, columns, data types, constraints, indexes, and views.

Result Cardinality - The number of rows returned as a result of executing an SQL query.

Query Execution Plan - The sequence of steps that the DBMS plans to execute to obtain the query result. Used to identify "bottlenecks".

# Functional Requirements

## Connection Management:

- The system must allow creating, editing, and deleting connections to SQLite database files.
- The system must verify the availability and integrity of the DB file upon connection.

## DataBase Schema Management (DDL):

- The system must provide a visual or text-based interface for creating, modifying, and deleting tables.
- The system must support constraint management (PRIMARY KEY, FOREIGN KEY, UNIQUE, CHECK, NOT NULL).
- The system must allow creating, modifying, and deleting indexes and views.
- The system must provide the ability to export the current DB schema as a single SQL script (DDL).

## Query Execution and Data Visualization:

- The system must allow entering and executing arbitrary SQL queries (SELECT, INSERT, UPDATE, DELETE, etc.).
- The system must display the results of SELECT queries as a table (Data Grid).
- For high result cardinality, the system must load and display data in chunks without loading the entire volume into RAM.
- The system must display general query statistics: execution time, number of affected rows, and total number of rows in the result (asynchronously).

## Query Plan Analysis and Visualization:

- The system must extract the query execution plan (e.g., using EXPLAIN QUERY PLAN in SQLite).
- The system must visualize the query plan as a tree or graph.
- The system must highlight potentially suboptimal operations (e.g., Full Table Scan).

**Architecture Scalability:**

- The architecture must allow adding support for other DBMSs (PostgreSQL, MySQL) in the future without rewriting the system core (using Strategy/Plugin patterns).

# Non-Functional Requirements

## Reliability:

- The system must not crash when invalid SQL syntax is entered or when working with corrupted DB files. Errors must be caught and displayed to the user.

## Usability:

- The interface must support SQL syntax highlighting, autocomplete, and dark/light themes.

# System Design

## Use cases for the first stage

This document describes the user's actions in the application for working with the relational database schema, executing SQL-queries, and analyzing their execution plans

In the first stage, SQLite is considered; support for PostgreSQL, MySQL, and collaboration relate to the possible further development of the project

## Actors and boundaries

The main actor is the Database Browser user, who connects the database file and works with its contents and schema

Accounts, registration, and administrator roles are not included in the requirements of the first stage, so the same user type is specified for all scenarios.

Prerequisites, successful flow, unsuccessful and alternative flows, as well as a postcondition are given for each scenario separately

## Connecting and viewing the database

### UC-01 Connect an SQLite database

|                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Actor                              | Database Browser User                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Goal                               | To open the SQLite file and start working with the database                                                                                                                                                                                                                                                                                                                                                                                 |
| Prerequisites                      | The application is running                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Successful flow                    | <ol style="list-style-type: none"><li>1 The user selects a saved connection or specifies an SQLite file for a new connection</li><li>2 The system checks the availability of the file, its format, and the integrity of the database</li><li>3 The system opens the connection and displays the database in the interface</li><li>4 If the file is specified for the first time, the system saves a record connections to re-open</li></ol> |
| Unsuccessful flow and alternatives | <p>The user deselects the file-the connection is not being created</p> <p>The file is missing, unavailable, corrupted, or is not an SQLite database - the system reports the reason and does not open the connection</p>                                                                                                                                                                                                                    |
| Postcondition on success           | The connection is open, the database is available for viewing and executing queries                                                                                                                                                                                                                                                                                                                                                         |

## UC-02 View Database Schema

|                                     |                                                                                                                                                                                                                                                                                                           |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actor</b>                        | Database Browser User                                                                                                                                                                                                                                                                                     |
| <b>Goal</b>                         | To find out the composition and structure of open database objects                                                                                                                                                                                                                                        |
| <b>Prerequisites</b>                | SQLite database is connected                                                                                                                                                                                                                                                                              |
| <b>Successful flow</b>              | <b>1</b> User selects the connected database<br><b>2</b> System shows tables, indexes, and views<br><b>3</b> User selects a schema object<br><b>4</b> System shows its definition: for a table-columns, types, and constraints; for an index-a table and indexed columns; for the view - defining request |
| <b>Failed flow and alternatives</b> | There are no objects in the database - the system shows an empty list<br><br>Object information could not be read - the system reports an error and saves the connection                                                                                                                                  |
| <b>Postcondition on success</b>     | The User sees the structure of the selected object, the database is not changed                                                                                                                                                                                                                           |

## Managing the database schema

### UC-03 Create a Table

|                                           |                                                                                                                                                                                                                                                                                  |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actor</b>                              | Database Browser User                                                                                                                                                                                                                                                            |
| <b>Goal</b>                               | To add a table with specified columns and constraints                                                                                                                                                                                                                            |
| <b>Prerequisites</b>                      | SQLite database is connected and writable                                                                                                                                                                                                                                        |
| <b>Successful Flow</b>                    | <b>1</b> User sets the table name, columns, and data types<br><b>2</b> If necessary, the user sets PRIMARY KEY, FOREIGN KEY, UNIQUE, CHECK, and NOT NULL<br><b>3</b> User confirms creation<br><b>4</b> The system passes the SQLite definition and updates the displayed schema |
| <b>Unsuccessful flow and alternatives</b> | The user cancels creation - the schema does not change<br><br>The name is occupied, the definition is incorrect, or the entry is forbidden - the system reports an error, the table is not created                                                                               |
| <b>Postcondition on success</b>           | The table is created and displayed in the schema                                                                                                                                                                                                                                 |

## UC-04 Change Table Structure

|                                           |                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actor</b>                              | Database Browser User                                                                                                                                                                                                                                                                                                                                    |
| <b>Goal</b>                               | To change columns or constraints of an existing table                                                                                                                                                                                                                                                                                                    |
| <b>Prerequisites</b>                      | SQLite database is connected and writable, the table exists                                                                                                                                                                                                                                                                                              |
| <b>Successful flow</b>                    | <b>1</b> User selects the table and sets the required change in its structure<br><b>2</b> System checks the change's applicability to the current schema and data<br><b>3</b> User confirms the change<br><b>4</b> System applies the entire change and updates the displayed schema                                                                     |
| <b>Unsuccessful flow and alternatives</b> | If the user cancels the change - the table is saved in the same form<br><br>The change is not possible using SQLite, violates restrictions or conflicts with data - the system reports the reason and retains the same structure<br><br>An error occurred during application - the system rolls back the change and shows the actual state of the schema |
| <b>Postcondition on success</b>           | The table structure corresponds to the saved definition                                                                                                                                                                                                                                                                                                  |

## UC-05 Delete Table

|                                           |                                                                                                                                                                                                                                                   |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actor</b>                              | User Database Browser                                                                                                                                                                                                                             |
| <b>Goal</b>                               | Delete the table along with its data                                                                                                                                                                                                              |
| <b>Prerequisites</b>                      | SQLite database is connected and writable, the table exists                                                                                                                                                                                       |
| <b>Successful flow</b>                    | <b>1</b> User selects the table to delete<br><b>2</b> The system warns about deleting the table and its rows and requests confirmation<br><b>3</b> The user confirms the deletion<br><b>4</b> The system deletes the table and updates the schema |
| <b>Unsuccessful flow and alternatives</b> | The User refuses from deletion - the table and data are saved<br>SQLite cannot delete the table - the system reports the reason and shows the actual state of the database                                                                        |
| <b>Postcondition on success</b>           | The table and its data are deleted from the database                                                                                                                                                                                              |

## UC-06 Create an Index

|              |                       |
|--------------|-----------------------|
| <b>Actor</b> | Database Browser User |
|--------------|-----------------------|

|                                           |                                                                                                                                                                                                                                         |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Goal</b>                               | To create an index for selected table columns                                                                                                                                                                                           |
| <b>Prerequisites</b>                      | SQLite database is connected and writable, the table exists                                                                                                                                                                             |
| <b>Successful flow</b>                    | <b>1</b> User sets the index name and one or more table columns<br><b>2</b> If necessary, the user specifies the uniqueness attribute<br><b>3</b> User confirms creation<br><b>4</b> The system creates an index and updates the schema |
| <b>Unsuccessful flow and alternatives</b> | User cancels creation-the schema does not change<br>The name is occupied, the column does not exist, or the available data violates the uniqueness requirement - the system reports an error, the index is not created                  |
| <b>Postcondition on success</b>           | The Index is created and displayed in the schema                                                                                                                                                                                        |

### UC-07 Change the Index

|                                           |                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actor</b>                              | User Database Browser                                                                                                                                                                                                                                                                                                                         |
| <b>Goal</b>                               | To change the definition of the user-created index                                                                                                                                                                                                                                                                                            |
| <b>Prerequisites</b>                      | SQLite database is connected and writable, the user index exists                                                                                                                                                                                                                                                                              |
| <b>Successful flow</b>                    | <b>1</b> The user selects the index and sets a new set of columns or a unique attribute<br><b>2</b> The system checks the new definition<br><b>3</b> The user confirms the change<br><b>4</b> The system replaces the index in one operation and updates the schema                                                                           |
| <b>Unsuccessful flow and alternatives</b> | The user cancels the change - the previous index is saved<br>The definition is incorrect or the data violates the uniqueness requirement - the system reports an error and saves the previous index<br>The index created by the PRIMARY KEY or UNIQUE constraint is selected - the system suggests changing the table structure via the UC-04 |
| <b>Postcondition on success</b>           | The User index has a new definition                                                                                                                                                                                                                                                                                                           |

### UC-08 Delete Index

|                      |                                                              |
|----------------------|--------------------------------------------------------------|
| <b>Actor</b>         | User Database Browser                                        |
| <b>Goal</b>          | Delete user-created index                                    |
| <b>Prerequisites</b> | SQLite database is connected and writable, user index exists |

|                                           |                                                                                                                                                                                                            |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Successful flow</b>                    | <b>1</b> User selects index<br><b>2</b> System shows linked table and asks for confirmation<br><b>3</b> User confirms deletion<br><b>4</b> System deletes index and updates schema                         |
| <b>Unsuccessful flow and alternatives</b> | User cancels action-index is saved<br>Index cannot be deleted separate from the table constraint, or SQLite rejects the operation - the system reports the reason and saves the actual state of the schema |
| <b>Postcondition on success</b>           | The index is deleted, the table rows are saved                                                                                                                                                             |

### UC-09 Create a View

|                                           |                                                                                                                                                                |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actor</b>                              | Database Browser User                                                                                                                                          |
| <b>Goal</b>                               | To add a named view based on SELECT to the schema                                                                                                              |
| <b>Prerequisites</b>                      | SQLite database is connected and writable                                                                                                                      |
| <b>Successful flow</b>                    | <b>1</b> User sets the view name and defining SELECTquery<br><b>2</b> User confirms creation<br><b>3</b> System creates the view and updates the schema        |
| <b>Unsuccessful flow and alternatives</b> | User cancels creation-schema does not change<br>Name is occupied or SQLite is used rejects the definition - the system shows an error, the view is not created |
| <b>Postcondition on success</b>           | The view is created and displayed in the schema                                                                                                                |

### UC-10 Change the view

|                                           |                                                                                                                                                                                                                                 |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actor</b>                              | Database Browser User                                                                                                                                                                                                           |
| <b>Goal</b>                               | To change the name or defining query of the view                                                                                                                                                                                |
| <b>Prerequisites</b>                      | SQLite database is connected and writable, the view exists                                                                                                                                                                      |
| <b>Successful flow</b>                    | <b>1</b> User selects the view and views its current definition<br><b>2</b> User sets a new name or SELECTquery and confirms the change<br><b>3</b> System replaces the definition as a single operation and updates the schema |
| <b>Unsuccessful flow and alternatives</b> | The user cancels editing - the view does not change<br>SQLite rejects the new definition or the replacement does not complete - the system reports an error and saves the previous view                                         |
| <b>Postcondition</b>                      | Next subsequent accesses to the view use the saved definition                                                                                                                                                                   |

|                   |  |
|-------------------|--|
| <b>on success</b> |  |
|-------------------|--|

### UC-11 Delete a View

|                                           |                                                                                                                                                                                 |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actor</b>                              | User Database Browser                                                                                                                                                           |
| <b>Goal</b>                               | To remove a view from the schema                                                                                                                                                |
| <b>Prerequisites</b>                      | SQLite database is connected and writable, the view exists                                                                                                                      |
| <b>Successful flow</b>                    | <b>1</b> User selects the view to delete<br><b>2</b> System requests confirmation<br><b>3</b> User confirms deletion<br><b>4</b> System deletes the view and updates the schema |
| <b>Unsuccessful flow and alternatives</b> | User cancels the action - the view is saved<br>SQLite rejects deletion - system reports error and shows the actual state of the schema                                          |
| <b>Postcondition on success</b>           | The view is deleted, data from the source tables is saved                                                                                                                       |

### UC-12 Export Database Schema

|                                           |                                                                                                                                                                                                                                                                              |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actor</b>                              | Database Browser User                                                                                                                                                                                                                                                        |
| <b>Goal</b>                               | To get a single SQLscript with the definition of the current schema                                                                                                                                                                                                          |
| <b>Prerequisites</b>                      | SQLite database is connected, the schema is readable                                                                                                                                                                                                                         |
| <b>Successful flow</b>                    | <b>1</b> The user selects the schema export and specifies the path to the SQLfile<br><b>2</b> The system collects DDL tables with constraints, indexes, and views<br><b>3</b> The system saves a single script and reports successful export                                 |
| <b>Unsuccessful flow and alternatives</b> | The user cancels saving - no new file is created<br>The file already exists - the system asks for confirmation of replacement<br>The schema could not be read or the file could not be written - the system reports the reason, incomplete export is not issued as completed |
| <b>Postcondition on success</b>           | The SQL-file contains definitions of schema objects without data strings                                                                                                                                                                                                     |

## Queries and analysis

### UC-13 Execute an SQLQuery

|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actor</b>                              | Database Browser User                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Goal</b>                               | To Execute SQL and get rows or result of changing data or schema                                                                                                                                                                                                                                                                                                                                                       |
| <b>Prerequisites</b>                      | SQLite database is connected                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Successful flow</b>                    | <b>1</b> User enters an SQLquery and runs it<br><b>2</b> The system executes the query without blocking the interface<br><b>3</b> For a query with returned rows, the system shows a table result, for changing data or schema - the result of execution<br><b>4</b> The System shows available metrics: execution time and the number of affected rows, if applicable; after changing the schema, updates its display |
| <b>Unsuccessful flow and alternatives</b> | The User cancels input before starting-the database does not change<br>SQL contains an error, accesses a missing object, or SQLite rejects the operation-the system shows the reason and remains functional<br>The query did not return any rows - the system shows empty result without an error message                                                                                                              |
| <b>Postcondition on success</b>           | , the request is completed, the user sees its result and available statistics                                                                                                                                                                                                                                                                                                                                          |

### UC-14 View Large Query Result

|                              |                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actor</b>                 | Database Browser User                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Goal</b>                  | To examine parts of a large SELECT result and estimate its size                                                                                                                                                                                                                                                                                                                            |
| <b>Prerequisites</b>         | Executed a UC-13 query returning a large set of rows                                                                                                                                                                                                                                                                                                                                       |
| <b>Successful flow</b>       | <b>1</b> The system shows the first batch of rows<br><b>2</b> The user requests the next batch or scrolls through the result<br><b>3</b> The system loads the necessary rows without loading the entire result into memory<br><b>4</b> The System calculates the total number of rows separately and shows it after the calculation is completed along with the available query statistics |
| <b>Unsuccessful flow and</b> | The user stops viewing - the displayed rows remain available<br>Upload the next batch failed - the system reports an error and                                                                                                                                                                                                                                                             |

|                                 |                                                                                                                                                                         |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>alternatives</b>             | saves the already shown rows<br><br>Counting the total number of rows failed - the system reports that the indicator is not available, without hiding the result itself |
| <b>Postcondition on success</b> | The User viewed required portions of the result and got the available statistics                                                                                        |

## UC-15 View Request Execution Plan

|                                           |                                                                                                                                                                                                                                                                             |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actor</b>                              | Database Browser User                                                                                                                                                                                                                                                       |
| <b>Purpose</b>                            | To understand the order of operations and use of indexes when executing SQL                                                                                                                                                                                                 |
| <b>Prerequisites</b>                      | SQLite database is connected, the user entered a query for analysis                                                                                                                                                                                                         |
| <b>Successful flow</b>                    | <b>1</b> The user requests a plan for the entered SQL<br><b>2</b> The system receives the SQLite plan<br><b>3</b> The system shows the steps of the plan in the form of a tree or graph<br><b>4</b> The System marks operations for verification, such as a full table scan |
| <b>Unsuccessful flow and alternatives</b> | The request is incorrect or the plan was not received - the system shows the reason and saves the entered SQL                                                                                                                                                               |
| <b>Postcondition if successful</b>        | The User sees the plan and can evaluate the selected SQLite operations without changing the database data                                                                                                                                                                   |

## Saved connections

### UC-16 Change a saved connection

|                              |                                                                                                                                                                                                                                                                                                                    |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actor</b>                 | Database Browser User                                                                                                                                                                                                                                                                                              |
| <b>Goal</b>                  | To update the parameters of the connection record to the SQLite file                                                                                                                                                                                                                                               |
| <b>Prerequisites</b>         | There is a saved connection record in the application                                                                                                                                                                                                                                                              |
| <b>Successful flow</b>       | <b>1</b> The user selects the connection record for editing<br><b>2</b> The system shows the current path to the file<br><b>3</b> The user specifies a new path and saves changes<br><b>4</b> The System checks the availability, format, and integrity of the selected database and updates the connection record |
| <b>Unsuccessful flow and</b> | If the User cancels editing - the record does not change                                                                                                                                                                                                                                                           |

|                                 |                                                                                                                     |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>alternatives</b>             | A new file is missing, unavailable, or incorrect - the system reports the reason, the previous parameters are saved |
| <b>Postcondition on success</b> | The connection record points to the verified SQLite file                                                            |

### UC-17 Delete a saved connection

|                                           |                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actor</b>                              | Database Browser User                                                                                                                                                                                                                                                                                                                                               |
| <b>Goal</b>                               | To remove an unnecessary connection record from                                                                                                                                                                                                                                                                                                                     |
| <b>Prerequisites</b>                      | There is a saved connection record                                                                                                                                                                                                                                                                                                                                  |
| <b>Successful flow</b>                    | <ol style="list-style-type: none"> <li><b>1</b> The user selects the connection record to delete</li> <li><b>2</b> The system explains that the database file will not be deleted and requests confirmation</li> <li><b>3</b> The user confirms the deletion</li> <li><b>4</b> The System closes the open connection and deletes its entry from the list</li> </ol> |
| <b>Unsuccessful flow and alternatives</b> | <p>When the user cancels the deletion - the record and the open connection are saved</p> <p>The connection failed to close or the record failed to delete - the system reports the reason and shows the actual state</p>                                                                                                                                            |
| <b>Postcondition on success</b>           | The connection record is deleted from the application, the SQLite file is saved to disk                                                                                                                                                                                                                                                                             |